L27: Wrapping up and structured review

Objectives

- revisit the original goals of the course and check in
- suggest some evidence based strategies for the final exam
- build a decision tree for everything in part III
- work through examples where the job is to choose the test

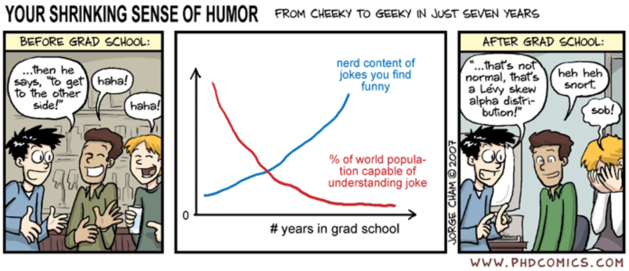
General wrap up

Goals for the semester

In addition to the learning objectives listed in your syllabus, our overarching goals for the semester are to develop:

- your ability to critically assess statistical information presented to you in scientific and non-scientific fora
- your sense of how to approach answering real world questions with data
- your ability to concisely and accurately describe statistical methods and results

Goals for the semester



***footnote:** *Thanks to Daragh at George Mason U. for this comic idea.*

Day 1 argument: this is a relevant class

I hoped to convince everyone here that statistics is relevant to everyone.

You make many decisions during your day that are influenced by statistics.

Statistics is not just relevant for public health, but also for other professions, including policy, journalism and law.

As we have tried to illustrate via the recurring "statistics is everywhere" segments, statistics is useful for understanding the news and the world around us.

Non-statistical considerations

Statistics tells us about the role of chance, that is, what would happen over many repeated samples.

We also want to think about the quality of the study design, what population was studied, whether the results are generalizable, and sources of potential bias.

In our interpretation we want to think about more than just the statistical result:

- how meaningful is the effect?
- if the results suggest a relationship, what are the risks and benefits of the exposure or treatment?
- what are the costs or implications of changing recommendations or guidelines?

Interpreting evidence: a mask use study

Effectiveness of Face Mask or Respirator Use in Indoor Public Settings for Prevention of SARS-CoV-2 Infection, California, February to December 2021. Andrejko et al., MMWR, February 4, 2022.

Test negative design case control study with matched observations, comparing self reported mask use between cases and controls.

Any face mask or respirator use in indoor public settings was associated with significantly lower odds of a positive test result compared with never using one (aOR $= 0.51$; 95% CI $= 0.29$ to 0.93). Always using a face mask or respirator was associated with lower adjusted odds of a positive test result compared with never wearing one (aOR $= 0.44$; 95% CI $= 0.24$ to 0.82).

Interpreting evidence: the results table

TABLE 2. Face mask or respirator use in indoor public settings among persons with positive and negative SARS-CoV-2 test results — California, February–December 2021

	SARS-CoV-2 infection status, no. (%)		Odds ratio (95% CI)	
Mask type and use*	**Positive (case-participant) N = 652**	**Negative (control-participant) N = 1,176**	**Unadjusted[†] [p-value]**	**Adjusted[§] [p-value]**
None (Ref)	44 (6.7)	42 (3.6)	—	—
Any use[†]	608 (93.3)	1,134 (96.4)	0.57 (0.37–0.90) [0.02]	0.51 (0.29–0.93) [0.03]
Some of the time	62 (9.5)	76 (6.5)	0.81 (0.47–1.41) [0.49]	0.71 (0.35–1.46) [0.36]
Most of the time	153 (23.5)	239 (20.3)	0.64 (0.40–1.05) [0.08]	0.55 (0.29–1.05) [0.07]
All of the time	393 (60.3)	819 (69.6)	0.49 (0.31–0.78) [<0.01]	0.44 (0.24–0.82) [<0.01]

Abbreviation: Ref = referent group.

* Trained interviewers administered a structured telephone-based questionnaire and asked participants to indicate whether they attended indoor public spaces during the 2 weeks before seeking a SARS-CoV-2 test. Participants who indicated attending these settings were further asked to specify whether they typically wore a face mask or respirator all, most, some, or none of the time while in these settings.

† Conditional logistic regression models were used to estimate the unadjusted odds of mask use by type of face mask or respirator worn in indoor public settings during the 2 weeks before testing. Models included matching strata defined by (for the period before June 15, 2021) the reopening tier of California in the county of residence and the week of SARS-CoV-2 testing.

§ Conditional logistic regression models were used to estimate the odds of face mask or respirator use in indoor public settings during the 2 weeks before testing, adjusting for COVID-19 vaccination status, household income, race/ethnicity, age group, sex, state region, and county population density. All models included matching strata defined by (for the period before June 15, 2021) the reopening tier of California in the county of residence, and the week of SARS-CoV-2 testing. To understand the effects of masking in community settings, this analysis was restricted to a subset of persons who did not indicate a known or suspected exposure to a SARS-CoV-2 case within 14 days of seeking a SARS-CoV-2 test. Adjusted models used a complete case analysis (454 case-participants and 789 control-participants). A sensitivity analysis using multiple imputation of missing covariate values obtained results similar to those reported in the table: adjusted odds ratios were 0.54 (95% CI = 0.33–0.89) for any mask use, 0.44 (95% CI = 0.27–0.73) for mask use all of the time, 0.62 (95% CI = 0.37–1.04) for mask use most of the time, and 0.77 (95% CI = 0.43–1.40) for mask use some of the time. An additional sensitivity analysis was conducted with additional adjustment for the reasons for SARS-CoV-2 testing as listed in Table 1 (experiencing symptoms, testing required for medical procedure, routine screening through work or school, pre-travel test, just wanted to see if I was infected, test required for admission to an event or gathering). The adjusted odds ratio was 0.42 (95% CI = 0.20–0.89) for any mask use as compared to no mask use upon additional adjustment for testing indications.

Evidence based suggestions: longhand notes

Pam A. Mueller and Dan Oppenheimer, Psychological Science 2014, Vol. 25(6) 1159-1168

The Pen Is Mightier Than the Keyboard: Advantages of Longhand Over Laptop Note Taking

Pam A. Mueller[1] and Daniel M. Oppenheimer[2]
[1]Princeton University and [2]University of California, Los Angeles

Evidence based suggestions: longhand notes

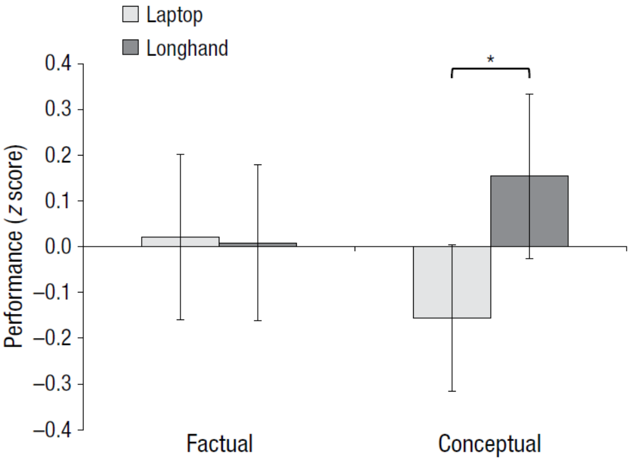


Fig. 1. Mean z-scored performance on factual-recall and conceptual-application questions as a function of note-taking condition (Study 1). The asterisk indicates a significant difference between conditions ($p < .05$). Error bars indicate standard errors of the mean.

Evidence based suggestions: anxiety reduction

Virginia Clinton and Stacy Meester, Teaching of Psychology 2019, Vol. 46(1) 92-95

A Comparison of Two In-Class Anxiety Reduction Exercises Before a Final Exam

Teaching of Psychology
2019, Vol. 46(1) 92-95

DOI: 10.1177/0098628318816182
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Virginia Clinton[1] and Stacy Meester[2]

Evidence based suggestions: anxiety reduction

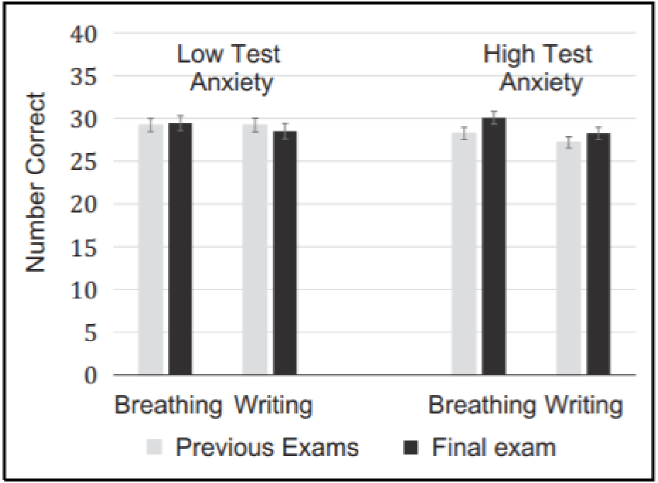


Figure 1. Previous exam and final exam performance by condition and level of trait test anxiety (means and $\pm$ 1 *SE*).

Overview of Part III

Part III: the basic goal

Our overarching goal in part III is really to take our two "recipes" for statistical inference

1) Hypothesis testing
2) Confidence intervals

and figure out which ingredients to add in different situations.

Choose the ingredients: four questions

We want to answer the questions:

1) What kind of an outcome variable are we working with?
2) How many groups or categories do we have data from that we want to compare?
3) Are the groups independent from each other, or are observations inherently related and therefore paired?
4) Do we meet the assumptions for a parametric test?

Choose the ingredients: three ingredients

Based on the answers to the previous questions, we choose our “ingredients”

1) the effect or difference we want to examine
2) the variability we have in the data
3) a distribution we will use to draw a critical value from, based on
   - our desired alpha
   - a one or two tailed hypothesis

Decision tree: continuous outcomes

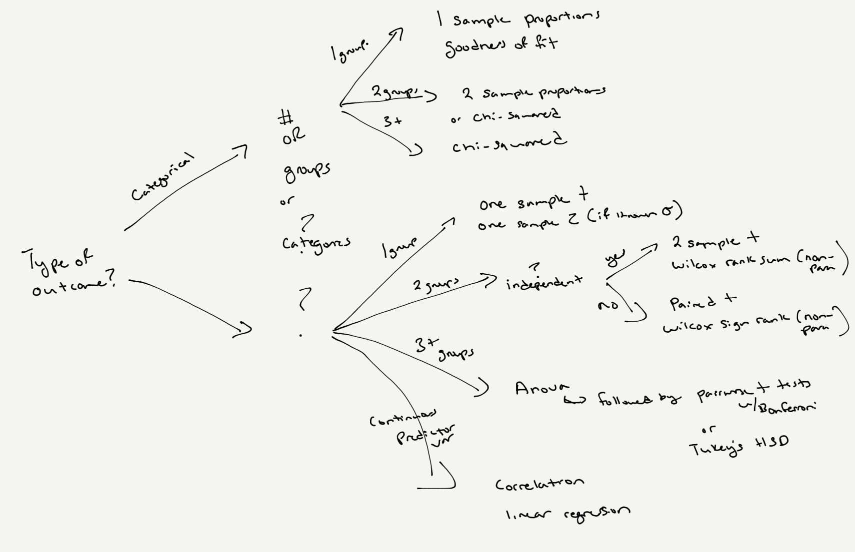
- One group, compared to a hypothesized value
  - one sample t, or one sample z if σ is known
- Two groups, independent
  - two sample t, or Wilcoxon rank sum if the assumptions fail
- Two groups, paired
  - paired t, or Wilcoxon signed rank if the assumptions fail
- Three or more groups
  - ANOVA, followed by pairwise t tests with a Bonferroni correction, or Tukey's HSD
- A continuous predictor
  - correlation, or linear regression

Decision tree: categorical outcomes

- One group, two categories
  - one sample z test for a proportion
- One group, three or more categories
  - chi-square goodness of fit
- Two groups, two categories
  - two sample z test for proportions, or a chi-square test of independence. On a 2x2 table these are the same test, and $\chi^2 = z^2$
- Two categorical variables, any size table
  - chi-square test of independence

Linear regression also handles categorical predictors, and there are bootstrap and permutation approaches for all of these comparisons.

Decision tree: the whole map



note that linear regression can also be done for categorical predictor variables and there are Bootstrap + Permutation approaches for all of these comparisons.

Recipe 1: hypothesis testing

1) Check the assumptions for the theoretical distribution
2) Create a ratio of the effect or difference to the variability in the data
3) Generate the probability of observing that difference or a more extreme one if the null is true
4) Make a decision to reject or not reject the null hypothesis

Recipe 2: confidence intervals

1) Generate your estimate
2) Calculate the critical value associated with your theoretical distribution
3) Multiply the critical value by the variability
4) Create your upper and lower bounds

For each test know:

- When to use it
- Any important assumptions that can be checked using data
- Appropriate visualization for the data
- What theoretical distribution we are using for inference
- How to construct the statistical test
    - what are the null and alternative hypotheses for the test
- How to construct the confidence interval
- Relevant syntax in R
- Any special notes or considerations

For example: the one sample t

One sample t

- used when we have 1 sample of a continuous outcome that we are comparing to a hypothesized value
- assuming an SRS, normality of the outcome, and independence of outcomes
- we might look at a histogram, a density plot or a qq plot
- compared to a t distribution with $n - 1$ degrees of freedom

For example: the one sample t

One sample t

- null: the mean is equal to the hypothesized mean
- alternative: could be one or two sided

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

For example: the one sample t

One sample t relevant syntax:

- `pt()` for the p-value from a t statistic
- `t.test(variable, alternative = "two.sided", mu = 0)`

Notes: think about when the test is robust to violations of the assumptions, and why.

Some examples - what tests?

Today's fun fact

There is an old claim that the length of your hand from your wrist to your elbow is the same as the length of your foot from your heel to your big toe.

If I wanted to show that these two measures are almost perfectly correlated, how might I do that?

You could run a correlation test, fit a linear regression, or show a scatterplot.

Today's fun fact: correlation is not agreement

Careful, “correlated” and “equal” are different claims.

A correlation test or a regression answers whether the two measures move together.

A paired t test answers a different question, namely whether the mean difference between them is zero. Two measures can be perfectly correlated and still differ by a constant.

What kind of plot would I expect to see? A scatterplot with points falling close to a straight line with a positive slope. If the two lengths really are equal, the points fall on the line $y = x$.

Example 1: staph infections

Researchers recruited 917 patients who had tested positive for *Staphylococcus aureus* and randomly assigned them to a staph-killing nasal ointment or placebo. They were interested in testing whether this drug was associated with a reduction in post-surgical infections. In the active treatment group 17 of 504 patients developed infections, in the placebo group 32 of 413 patients developed infections.

- What are the exposure and outcome variables?
- What kind of a test would you use for these data?
- What is the null hypothesis of this test?

Staph infections: the ingredients

1) the effect or difference we want to examine
2) the variability we have in the data
3) a distribution we will use to draw a critical value from, based on
    - our desired alpha
    - a one or two tailed hypothesis

Ingredient 1: the effect

Difference between two proportions:

$$\hat{p}_1 - \hat{p}_2 = \frac{17}{504} - \frac{32}{413} = 0.0337 - 0.0775 = -0.0438$$

Ingredient 2: variability

If the null hypothesis is true, then p_1 is truly equal to p_2. In that case our best estimate of the underlying proportion that they are both equal to is

$$\hat{p} = \frac{\text{no. successes in both samples}}{\text{no. individuals in both samples}} = \frac{17 + 32}{504 + 413} = 0.0534$$

Ingredient 2: variability

Our best guess at the SE for $\hat{p}_1 - \hat{p}_2$ under the null is

$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n_1} + \frac{\hat{p}(1-\hat{p})}{n_2}} = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

This is the SE for the difference between two proportions, with the pooled $\hat{p}$ substituted for both p_1 and p_2.

$$\sqrt{0.0534 \times 0.9466\left(\frac{1}{504} + \frac{1}{413}\right)} = 0.01493$$

Ingredient 3: the distribution

Here for two sample testing we have more than 10 successes and 10 failures in each group, so we feel comfortable using a normal approximation to the binomial.

We will use a Z distribution and an alpha of 0.05.

The test is one sided because we are only interested in the left side of the distribution, that is, in a decrease.

Staph infections: the test statistic

$$z = \frac{0.0337 - 0.0775}{\sqrt{0.0534 \times 0.9466\left(\frac{1}{504} + \frac{1}{413}\right)}} = -2.931$$

Staph infections: the p-value

```r
pnorm(-2.931)
```

```
## [1] 0.001689364
```

We are only interested in a reduction in infections, so we look only at the left tail of the distribution. That single tail is the p-value.

Staph infections: in R


```r
prop.test(x = c(17, 32),    # successes in each group
          n = c(504, 413), # sample sizes
          alternative = "less", correct = FALSE)
```

```
##
##  2-sample test for equality of proportions without
##  continuity correction
##
## data:  c(17, 32) out of c(504, 413)
## X-squared = 8.5906, df = 1, p-value = 0.00169
## alternative hypothesis: less
## 95 percent confidence interval:
##  -1.00000000 -0.01839005
## sample estimates:
##     prop 1     prop 2
## 0.03373016 0.07748184
```

Staph infections: the confidence interval

For the interval we do not pool, because we are no longer assuming the null is true.

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$-0.0438 \pm 1.96\sqrt{\frac{0.0337(1-0.0337)}{504} + \frac{0.0775(1-0.0775)}{413}} = -0.0438 \pm 0.0302$$

The 95% CI is -0.0740 to -0.0135. It excludes 0, which agrees with the small p-value.

Example 2: vitamin D and type 2 diabetes

August 8, 2019, Vitamin D Supplementation and Prevention of Type 2 Diabetes, New England Journal of Medicine.

BACKGROUND. Observational studies support an association between a low blood 25-hydroxyvitamin D level and the risk of type 2 diabetes. However, whether vitamin D supplementation lowers the risk of diabetes is unknown.

Example 2 cont: methods

METHODS. We randomly assigned adults who met at least two of three glycemic criteria for prediabetes (fasting plasma glucose level, 100 to 125 mg per deciliter; plasma glucose level 2 hours after a 75-g oral glucose load, 140 to 199 mg per deciliter; and glycated hemoglobin level, 5.7 to 6.4%) and no diagnostic criteria for diabetes to receive 4000 IU per day of vitamin D3 or placebo, regardless of the baseline serum 25-hydroxyvitamin D level.

What type of study is this?

What type of variable is the predictor, and how many groups?

Example 2 cont: results

RESULTS. By month 24, the mean serum 25-hydroxyvitamin D level in the vitamin D group was 54.3 ng per milliliter (from 27.7 at baseline), as compared with 28.8 ng per milliliter in the placebo group (from 28.2 at baseline). After a median follow-up of 2.5 years, the primary outcome of diabetes occurred in 293 participants in the vitamin D group and 323 in the placebo group.

What kind of test would you use for the vitamin D level comparison? What about for the diabetes outcome?

Example 2: other considerations

Why might we not see the result we were expecting?

Per the discussion in the article:

"Because vitamin D supplements are used increasingly in the U.S. adult population, approximately 8 of 10 participants had a baseline serum 25-hydroxyvitamin D level that was considered to be sufficient according to current recommendations (≥ 20 ng per milliliter) to reduce the risk of many outcomes, including diabetes. The high percentage of participants with adequate levels of vitamin D may have limited the ability of the trial to detect a significant effect."

Example 3

A study on the effects of vaping classifies people as "never vapers", "occasional vapers" and "frequent vapers". You interview a sample of 150 people in each group and use a questionnaire to derive a quantitative score, between 0 and 100, on stress levels.

What kind of an outcome is this?

How many groups, and are they independent?

What test is appropriate here?

Example 4

The amygdala is a brain structure involved in the processing of memory of emotional reactions. Ten subjects were shown emotional video clips and non emotional video clips in random order. They then had their memory of the clips assessed. Recall accuracy was scored from 1 to 100.

What type of data do you have?

What kind of a test is appropriate?

Example 5

A random sample of 700 births from local records shows this distribution across the days of the week. Do these data give evidence that local births are not equally likely on all days of the week?

Day	Births
Monday	110
Tuesday	124
Wednesday	104
Thursday	94
Friday	112
Saturday	72
Sunday	84

What test would we use here, and what is the null hypothesis?

Example 5: the expected counts

Under the null, all seven days are equally likely, so each expected count is $700 \times \frac{1}{7} = 100$.

Day	Observed	Expected
Monday	110	100
Tuesday	124	100
Wednesday	104	100
Thursday	94	100
Friday	112	100
Saturday	72	100
Sunday	84	100

All expected counts are well above 5, so the conditions are met.

Example 5: the test

Degrees of freedom are $k - 1 = 7 - 1 = 6$.

```
births <- c(110, 124, 104, 94, 112, 72, 84)
chisq.test(births)
```

```
##
##  Chi-squared test for given probabilities
##
## data:  births
## X-squared = 19.12, df = 6, p-value = 0.003966
```

Saturday and Sunday do almost all of the work here.

Example 6

You have heard that the grading scale is harsher at UC Berkeley than at other California universities. You want to test this rumor with data. You have the letter grades of a random sample of 350 Berkeley students who took PH142, and data on the letter grade distribution for undergraduate statistics courses in general from a California wide survey.

What kind of outcome is this? How many groups or samples?

What test would you consider here?

Example 6: which test

This is a categorical outcome measured on one sample, and the null gives us the share we expect in each category from an outside source.

That is a chi-square goodness of fit test, not a test of independence. There is only one variable here.

Example 6: the expected counts

The California wide survey gives the distribution for intro statistics courses as A = 50%, B = 30%, C = 15%, F = 5%.

Grade	Observed	Expected
A	224	?
B	99	?
C	17	?
F	10	?
Total	350	350

Example 6: the statistic

Grade	Observed	Expected	$(O-E)^2/E$
A	224	175	13.72
B	99	105	0.34
C	17	52.5	24.00
F	10	17.5	3.21
Total	350	350	41.28

```
pchisq(41.28, df = 3, lower.tail = FALSE)
```

```
## [1] 5.703407e-09
```

Common issues or questions

Parametric vs non-parametric

What does it mean when I say non-parametric?

Which tests have we covered that are non-parametric?

Common issues

Method used, versus p-value and CI, versus conclusion.

Appropriate interpretation of a p-value and of a confidence interval.

The relationship between a confidence interval and a p-value.

Good luck

Three things to carry into the exam:

- Start from the data, not from the formula. What kind of outcome, how many groups, paired or independent. The test follows from those answers.
- Write the conclusion in the units of the problem. A p-value on its own is not a conclusion.
- Use the evidence we just looked at. Longhand notes and a few minutes of breathing beforehand are cheap and they help.

It has been a pleasure teaching you all.